IEOR 221

Homework 5

Module 6 Options – Binomial Trees (October 6, 2025)

1) A stock price is currently $100. Over each of the next two 6-month periods it is expected to go up by 10% or down by 10%. The risk-free interest rate is 8% per annum with continuous compounding. What is the value of a 1-year European call option with a strike price of $100? (Hull 13.11)

2) For the situation considered in Problem 1), what is the value of a 1-year European put option with a strike price of $100? Verify that the European call and European put prices satisfy put–call parity. (Hull 13.12)

3) The current price of a non-dividend-paying biotech stock is $140 with a volatility of 25%. The risk-free rate is 4%. For a 3-month time step:
(a) What is the percentage up movement?
(b) What is the percentage down movement?
(c) What is the probability of an up movement in a risk-neutral world?
(d) What is the probability of a down movement in a risk-neutral world?
Use a two-step tree to value a 6-month European call option and a 6-month European put option. In both cases the strike price is $150. (Hull 13.24)

4) In Problem 3), suppose a trader sells 10,000 European call options and the two-step tree describes the behavior of the stock. How many shares of the stock are needed to hedge the 6-month European call for the first and second 3-month period? For the second time period, consider both the case where the stock price moves up during the first period and the case where it moves down during the first period. (Hull 13.25)